

Assignment 7 — Write-up

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Part A — Kernel Methods

A.1 Norm-based kernel bound (5 pts)

Claim.

$$\hat{\mathcal{R}}_S(\mathcal{H}_{K,B}) \leq \frac{B}{n} \sqrt{\sum_{i=1}^n K(x_i, x_i)}.$$

Proof.

Starting from the definition:

$$\hat{\mathcal{R}}_S(\mathcal{H}_{K,B}) = \mathbb{E}_\sigma \sup_{\|w\| \leq B} \frac{1}{n} \sum_{i=1}^n \sigma_i \langle w, \varphi(x_i) \rangle = \mathbb{E}_\sigma \sup_{\|w\| \leq B} \frac{1}{n} \left\langle w, \sum_{i=1}^n \sigma_i \varphi(x_i) \right\rangle.$$

The inner product $\langle w, v \rangle$ over $\|w\| \leq B$ is maximized by $w = B v / \|v\|$ (when $v \neq 0$), giving value $B\|v\|$. Hence:

$$\hat{\mathcal{R}}_S(\mathcal{H}_{K,B}) = \frac{B}{n} \mathbb{E}_\sigma \left\| \sum_{i=1}^n \sigma_i \varphi(x_i) \right\|.$$

By Jensen's inequality ($\|\cdot\|$ is concave under $\mathbb{E}$ via $\mathbb{E}[\|v\|] \leq \sqrt{\mathbb{E}[\|v\|^2]}$):

$$\hat{\mathcal{R}}_S(\mathcal{H}_{K,B}) \leq \frac{B}{n} \sqrt{\mathbb{E}_\sigma \left\| \sum_{i=1}^n \sigma_i \varphi(x_i) \right\|^2}.$$

Expanding the squared norm:

$$\mathbb{E}_\sigma \left\| \sum_i \sigma_i \varphi(x_i) \right\|^2 = \sum_{i=1}^n \sum_{j=1}^n \mathbb{E}[\sigma_i \sigma_j] \langle \varphi(x_i), \varphi(x_j) \rangle.$$

Since $\sigma_1, \dots, \sigma_n$ are independent uniform signs, $\mathbb{E}[\sigma_i \sigma_j] = \mathbf{1}[i = j]$. Therefore:

$$\mathbb{E}_\sigma \left\| \sum_i \sigma_i \varphi(x_i) \right\|^2 = \sum_{i=1}^n \|\varphi(x_i)\|^2 = \sum_{i=1}^n K(x_i, x_i).$$

Combining:

$$\hat{\mathcal{R}}_S(\mathcal{H}_{K,B}) \leq \frac{B}{n} \sqrt{\sum_{i=1}^n K(x_i, x_i)}.$$

Distribution-level bound. Taking expectation over $S \sim \mathcal{D}^n$:

$$\mathcal{R}_{\mathcal{D}^n}(\mathcal{H}_{K,B}) = \mathbb{E}_S \hat{\mathcal{R}}_S(\mathcal{H}_{K,B}) \leq \frac{B}{n} \mathbb{E}_S \sqrt{\sum_{i=1}^n K(x_i, x_i)}.$$

By Jensen (concavity of $\sqrt{\cdot}$):

$$\mathbb{E}_S \sqrt{\sum_{i=1}^n K(x_i, x_i)} \leq \sqrt{\mathbb{E}_S \sum_{i=1}^n K(x_i, x_i)} = \sqrt{n \mathbb{E}[K(X, X)]}.$$

Hence:

$$\boxed{\mathcal{R}_{\mathcal{D}^n}(\mathcal{H}_{K,B}) \leq B \sqrt{\frac{\mathbb{E}[K(X, X)]}{n}}}.$$

Key insight. The bound depends on the kernel's diagonal values $K(x, x) = \|\varphi(x)\|^2$, not on the ambient dimension of $\mathcal{U}$. Even for infinite-dimensional RKHS (e.g., Gaussian kernel), the Rademacher complexity is finite and controlled by B and the average feature-map norm.

A.2 Representer theorem (5 pts)

Claim. Any minimizer w^* of $J(w) = \frac{1}{n} \sum_{i=1}^n \text{loss}(\langle w, \varphi(x_i) \rangle; y_i) + \lambda \|w\|^2$ has the form $w^* = \sum_{j=1}^n \alpha_j \varphi(x_j)$.

Proof. Let $V = \text{span}\{\varphi(x_1), \dots, \varphi(x_n)\}$ and decompose any $w \in \mathcal{U}$ as $w = w_{\parallel} + w_{\perp}$ where $w_{\parallel} \in V$ and $w_{\perp} \in V^{\perp}$.

Since $w_{\perp} \perp \varphi(x_i)$ for all i :

$$\langle w, \varphi(x_i) \rangle = \langle w_{\parallel}, \varphi(x_i) \rangle + \langle w_{\perp}, \varphi(x_i) \rangle = \langle w_{\parallel}, \varphi(x_i) \rangle.$$

So the loss terms depend only on $w_{\parallel}$. Meanwhile, by Pythagoras:

$$\|w\|^2 = \|w_{\parallel}\|^2 + \|w_{\perp}\|^2 \geq \|w_{\parallel}\|^2,$$

with equality iff $w_{\perp} = 0$.

Therefore $J(w) = J_{\text{loss}}(w_{\parallel}) + \lambda(\|w_{\parallel}\|^2 + \|w_{\perp}\|^2) \geq J_{\text{loss}}(w_{\parallel}) + \lambda\|w_{\parallel}\|^2 = J(w_{\parallel})$. Since $\lambda > 0$, the inequality is strict whenever $w_{\perp} \neq 0$. So any minimizer must satisfy $w_{\perp} = 0$, i.e., $w^* \in V = \text{span}\{\varphi(x_1), \dots, \varphi(x_n)\}$. Hence $w^* = \sum_{j=1}^n \alpha_j \varphi(x_j)$ for some $\alpha \in \mathbb{R}^n$. $\square$

Rewriting J in terms of α and the Gram matrix. Substituting $w = \sum_j \alpha_j \varphi(x_j)$:

- **Predictions:**

$$\langle w, \varphi(x_i) \rangle = \sum_{j=1}^n \alpha_j \langle \varphi(x_j), \varphi(x_i) \rangle = \sum_{j=1}^n \alpha_j K(x_j, x_i) = (K_S \alpha)_i.$$

- **Norm:**

$$\|w\|^2 = \sum_{j,k} \alpha_j \alpha_k \langle \varphi(x_j), \varphi(x_k) \rangle = \alpha^\top K_S \alpha.$$

So the objective becomes:

$$J(\alpha) = \frac{1}{n} \sum_{i=1}^n \text{loss}((K_S \alpha)_i; y_i) + \lambda \alpha^\top K_S \alpha.$$

This is a finite-dimensional optimization over $\alpha \in \mathbb{R}^n$, in which the data enter only through the $n \times n$ Gram matrix K_S . The “kernel trick”: we never need to compute $\varphi(x)$ explicitly — only the pairwise kernel evaluations $K(x_i, x_j)$.

Part B — ℓ_1 Rademacher Complexity

B.1 Convex hulls (8 pts)

Claim. For any finite class $\mathcal{F}$ of real-valued functions, $\hat{\mathcal{R}}_S(\text{conv}(\mathcal{F})) = \hat{\mathcal{R}}_S(\mathcal{F})$.

Proof.

“ $\geq$ ” **direction.** Since $\mathcal{F} \subseteq \text{conv}(\mathcal{F})$, we have $\hat{\mathcal{R}}_S(\mathcal{F}) \leq \hat{\mathcal{R}}_S(\text{conv}(\mathcal{F}))$.

“ $\leq$ ” **direction.** We show the supremum over $\text{conv}(\mathcal{F})$ equals the supremum over $\mathcal{F}$, for each fixed σ .

Fix $\sigma = (\sigma_1, \dots, \sigma_n)$. Define $\Phi(h) = \frac{1}{n} \sum_{i=1}^n \sigma_i h(x_i)$, which is a linear functional of h . For any $h = \sum_{f \in \mathcal{F}} a_f f \in \text{conv}(\mathcal{F})$ (with $a_f \geq 0$, $\sum a_f = 1$):

$$\Phi(h) = \sum_{f \in \mathcal{F}} a_f \Phi(f) \leq \max_{f \in \mathcal{F}} \Phi(f).$$

The inequality follows because $\Phi(h)$ is a convex combination of $\{\Phi(f)\}_{f \in \mathcal{F}}$, which is at most the maximum. Therefore:

$$\sup_{h \in \text{conv}(\mathcal{F})} \Phi(h) = \max_{f \in \mathcal{F}} \Phi(f) = \sup_{f \in \mathcal{F}} \Phi(f).$$

Taking $\mathbb{E}_\sigma$ of both sides:

$$\hat{\mathcal{R}}_S(\text{conv}(\mathcal{F})) = \mathbb{E}_\sigma \sup_{h \in \text{conv}(\mathcal{F})} \Phi(h) = \mathbb{E}_\sigma \max_{f \in \mathcal{F}} \Phi(f) = \hat{\mathcal{R}}_S(\mathcal{F}).$$

□

Key insight. The supremum of a linear functional over a convex set is attained at an extreme point. The extreme points of $\text{conv}(\mathcal{F})$ are exactly the elements of $\mathcal{F}$, so Rademacher complexity cannot increase by taking convex hulls.

B.2 ℓ_1 linear predictors (10 pts)

Claim. $\hat{\mathcal{R}}_S(\mathcal{H}_B^1) \leq BR\sqrt{\frac{2\ln(2d)}{n}}$.

Proof. We reduce $\mathcal{H}_B^1$ to the convex hull of a finite class, then apply Massart's lemma.

Step 1: Decompose the ℓ_1 ball as a scaled convex hull.

The extreme points of the ℓ_1 ball $\{w : \|w\|_1 \leq B\}$ are $\{0, \pm Be_1, \dots, \pm Be_d\}$. Any w with $\|w\|_1 \leq B$ can be written as $w = \sum_{j=1}^d (|w_j|/B) \cdot B \text{sign}(w_j) e_j + (1 - \|w\|_1/B) \cdot 0$, a convex combination with non-negative weights summing to 1. Define the finite class:

$$\mathcal{F} = \{x \mapsto B\sigma\varphi_j(x) : j \in [d], \sigma \in \{-1, +1\}\} \cup \{0\}.$$

Then $\mathcal{H}_B^1 = \{x \mapsto \langle w, \varphi(x) \rangle : \|w\|_1 \leq B\} = \text{conv}(\mathcal{F})$.

Step 2: Apply B.1. By the convex-hull identity:

$$\hat{\mathcal{R}}_S(\mathcal{H}_B^1) = \hat{\mathcal{R}}_S(\text{conv}(\mathcal{F})) = \hat{\mathcal{R}}_S(\mathcal{F}).$$

Step 3: Apply Massart's finite-class lemma. For a finite class $\mathcal{F}$ with $|\mathcal{F}|$ elements, Massart's lemma states:

$$\hat{\mathcal{R}}_S(\mathcal{F}) \leq \frac{\max_{f \in \mathcal{F}} \|v_f\|_2}{n} \cdot \sqrt{2 \ln |\mathcal{F}|},$$

where $v_f = (f(x_1), \dots, f(x_n)) \in \mathbb{R}^n$ is the evaluation vector.

Here $|\mathcal{F}| = 2d + 1 \leq 2d$ (we can use $2d$ since the 0 function only reduces the max). For $f = B\sigma\varphi_j$:

$$\|v_f\|_2^2 = \sum_{i=1}^n B^2 \varphi_j(x_i)^2 \leq nB^2 R^2,$$

since $|\varphi_j(x_i)| \leq \|\varphi(x_i)\|_\infty \leq R$.

Therefore:

$$\hat{\mathcal{R}}_S(\mathcal{F}) \leq \frac{BR\sqrt{n}}{n} \sqrt{2 \ln(2d)} = BR\sqrt{\frac{2 \ln(2d)}{n}}.$$

$$\hat{\mathcal{R}}_S(\mathcal{H}_B^1) \leq BR\sqrt{\frac{2 \ln(2d)}{n}}.$$

The bound is **logarithmic in d** : the ℓ_1 constraint replaces the linear-in- d scaling (from VC dimension) with a $\sqrt{\log d}$ factor, at the cost of depending on the norm bound B and feature bound R .

B.3 Comparison with sparsity bounds (7 pts)

Rademacher-based uniform convergence for $\mathcal{H}_B^1$. Using the week 7 uniform convergence theorem for a G -Lipschitz loss ℓ bounded in $[0, 1]$: with probability $\geq 1 - \delta$ over $S \sim \mathcal{D}^n$, for all $\|w\|_1 \leq B$,

$$L_{\mathcal{D}}^{\ell}(w) - L_S^{\ell}(w) \leq 2G \hat{\mathcal{R}}_S(\mathcal{H}_B^1) + \sqrt{\frac{\ln(1/\delta)}{2n}}.$$

Substituting the B.2 bound:

$$L_{\mathcal{D}}^{\ell}(w) - L_S^{\ell}(w) \leq 2GBR \sqrt{\frac{2 \ln(2d)}{n}} + \sqrt{\frac{\ln(1/\delta)}{2n}}.$$

Comparison of the two bounds.

	VC / sparsity bound	ℓ_1 Rademacher bound
Complexity measure	Sparsity k (combinatorial)	ℓ_1 norm B (scale-sensitive)
Dimension dependence	$k \log(ed/k)$	$\log(2d)$
Loss functional	0/1 loss directly	Lipschitz surrogate ℓ
Bound	$C \sqrt{\frac{k \log(ed/k) + \log(1/\delta)}{n}}$	$O\left(GBR \sqrt{\frac{\log d}{n}} + \sqrt{\frac{\log(1/\delta)}{n}}\right)$

Scale-sensitivity. The ℓ_1 bound is scale-sensitive: the complexity is $B = \|w\|_1$, a continuous quantity. Two predictors with the same support but different coefficient magnitudes get different bounds. The VC bound is scale-blind: it depends only on the combinatorial sparsity k , treating $w = (1, 0, \dots, 0)$ and $w = (10^6, 0, \dots, 0)$ identically.

Extra step for 0/1 error. The ℓ_1 bound controls surrogate loss L^{ℓ} , not 0/1 loss directly. To say anything about 0/1 error, one needs a *calibration inequality* relating $L^{0/1}$ to L^{ℓ} (e.g., for the hinge loss, $L^{0/1} \leq L^{\text{hinge}}$, so the surrogate bound upper-bounds the 0/1 generalization gap). The sparsity bound controls 0/1 error directly.

A predictor where the sparsity bound is vacuous but the ℓ_1 bound is not. Consider $w = (1/d, 1/d, \dots, 1/d) \in \mathbb{R}^d$ — a *dense* predictor with $\|w\|_1 = 1$ and sparsity $k = d$. The VC/sparsity bound gives generalization gap $\sim \sqrt{d \log(ed/d)/n} = \sqrt{d/n}$, which is vacuous when $d \gg n$. The ℓ_1 bound gives $\sim \sqrt{\log(d)/n}$, which remains meaningful even for $d \gg n$, because $B = \|w\|_1 = 1$ is small despite full support. This is exactly the regime where ℓ_1 regularization (LASSO) outperforms subset selection: the predictor is not sparse, but it has small ℓ_1 norm.

Part C — Normalized ℓ_1 Margins and FixedBoost

C.1 Margin generalization (10 pts)

Setup. $S \sim \mathcal{D}^n$ i.i.d., $\mathcal{G}$ is a finite symmetric base class, $f_w(x) = \sum_{g \in \mathcal{G}} w_g g(x)$, and $\text{margin}_S(w) = \min_i y_i f_w(x_i) / \|w\|_1 \geq \gamma > 0$.

Goal. Bound $\Pr_{(x,y) \sim \mathcal{D}}[\text{sign}(f_w(x)) \neq y]$.

Step 1: Use the ramp surrogate. Define the ramp loss at scale γ :

$$\ell_\gamma(z) = \begin{cases} 1 & z \leq 0 \\ 1 - z/\gamma & 0 < z < \gamma \\ 0 & z \geq \gamma \end{cases}$$

This satisfies: (i) $\mathbf{1}[z < 0] \leq \ell_\gamma(z)$ for all z (it upper-bounds 0/1 loss), and (ii) ℓ_γ is $1/\gamma$ -Lipschitz, and (iii) ℓ_γ is bounded in $[0, 1]$.

For any w with $\text{margin}_S(w) \geq \gamma$, the normalized predictor $\bar{w} = w/\|w\|_1$ has $y_i f_{\bar{w}}(x_i) \geq \gamma$ for all i , so $\ell_\gamma(y_i f_{\bar{w}}(x_i)) = 0$ for all i . Thus $L_S^{\ell_\gamma}(\bar{w}) = 0$.

Step 2: Apply the B.3 uniform convergence bound. The predictor $f_{\bar{w}}$ has $\|\bar{w}\|_1 = 1$, so $\bar{w} \in \mathcal{H}_1^1$. The feature map here is $\varphi(x) = (g(x))_{g \in \mathcal{G}} \in \{-1, +1\}^{|\mathcal{G}|}$, so $\|\varphi(x)\|_\infty = 1$, i.e., $R = 1$. The Lipschitz constant of ℓ_γ is $G = 1/\gamma$. Applying B.3 with $B = 1$, $R = 1$, $G = 1/\gamma$, and dimension $|\mathcal{G}|$:

With probability $\geq 1 - \delta$, for all $\|\bar{w}\|_1 \leq 1$:

$$L_{\mathcal{D}}^{\ell_\gamma}(\bar{w}) - L_S^{\ell_\gamma}(\bar{w}) \leq \frac{2}{\gamma} \sqrt{\frac{2 \ln(2|\mathcal{G}|)}{n}} + \sqrt{\frac{\ln(1/\delta)}{2n}}.$$

Step 3: Combine. Since $L_S^{\ell_\gamma}(\bar{w}) = 0$ and $L_{\mathcal{D}}^{0/1}(\bar{w}) \leq L_{\mathcal{D}}^{\ell_\gamma}(\bar{w})$:

$$\boxed{\Pr_{(x,y) \sim \mathcal{D}} [\text{sign}(f_w(x)) \neq y] \leq \frac{2}{\gamma} \sqrt{\frac{2 \ln(2|\mathcal{G}|)}{n}} + \sqrt{\frac{\ln(1/\delta)}{2n}}.}$$

The bound depends logarithmically on $|\mathcal{G}|$ and polynomially on $1/\gamma$: a large normalized ℓ_1 margin implies good generalization.

C.2 FixedBoost implication (8 pts)

Given (black box). If every distribution D over the training sample admits a base predictor with weighted error $\leq \frac{1}{2} - \frac{\gamma}{2}$, then after $T = O(\log(n)/\gamma^4)$ rounds, FixedBoost returns w_T with:

$$\min_i y_i f_{w_T}(x_i) \geq \frac{\gamma}{2} \|w_T\|_1.$$

This means $\text{margin}_S(w_T) \geq \gamma/2$.

Generalization guarantee. Apply C.1 with margin $\gamma/2$ in place of γ . With probability $\geq 1 - \delta$ over $S \sim \mathcal{D}^n$:

$$\boxed{\Pr_{(x,y) \sim \mathcal{D}} [\text{sign}(f_{w_T}(x)) \neq y] \leq \frac{4}{\gamma} \sqrt{\frac{2 \ln(2|\mathcal{G}|)}{n}} + \sqrt{\frac{\ln(1/\delta)}{2n}}.}$$

The bound is meaningful (say $\leq \varepsilon$) when $n = \tilde{\Omega}(\log |\mathcal{G}|/(\gamma^2 \varepsilon^2))$. The number of FixedBoost rounds $T = O(\log(n)/\gamma^4)$ does not appear in the generalization bound — only the achieved margin $\gamma/2$ and the base-class size $|\mathcal{G}|$ matter.

Interpretation. FixedBoost converts a weak-learning assumption (edge $\gamma/2$ under every reweighting) into a classifier with normalized ℓ_1 margin $\geq \gamma/2$, which then translates into a generalization guarantee via the Rademacher bound. The round count T is the price paid in computation, not in statistical complexity.

C.3 Weak learning and normalized margin (8 pts)

Setup. The matrix $A \in \{-1, +1\}^{n \times |\mathcal{G}|}$ has $A_{ig} = y_i g(x_i)$. We show:

$$\max_w \text{margin}_S(w) = \text{best weak-learning edge.}$$

Definitions. Let γ_{WL}^* be the best weak-learning edge: the largest γ such that for every distribution $D \in \Delta_n$ over the n examples, some $g \in \mathcal{G}$ has edge $\geq \gamma$. The edge of g under D is:

$$\text{edge}_D(g) = \frac{1}{2} - \sum_i D_i \mathbf{1}[g(x_i) \neq y_i] = \frac{1}{2} \sum_i D_i \cdot A_{ig}.$$

So: $\gamma_{\text{WL}}^* = \max\{\gamma : \min_{D \in \Delta_n} \max_{g \in \mathcal{G}} \frac{1}{2} \sum_i D_i A_{ig} \geq \gamma\} = \min_{D \in \Delta_n} \max_{g \in \mathcal{G}} \frac{1}{2} \sum_i D_i A_{ig}$.

Equivalently, $\gamma_{\text{WL}}^* = \frac{1}{2} \min_{D \in \Delta_n} \max_{g \in \mathcal{G}} D^\top A_{\cdot g}$, where $A_{\cdot g}$ is the g -th column.

The normalized ℓ_1 margin as a max-min. For the margin side, using symmetry ($-g \in \mathcal{G}$ whenever $g \in \mathcal{G}$), we can assume WLOG that $w_g \geq 0$ for all g (by absorbing signs into the symmetric class). Then $\|w\|_1 = \sum_g w_g$, and:

$$\text{margin}_S(w) = \min_i \frac{y_i f_w(x_i)}{\|w\|_1} = \min_i \frac{\sum_g w_g A_{ig}}{\sum_g w_g} = \min_i \sum_g p_g A_{ig},$$

where $p = w/\|w\|_1 \in \Delta_{|\mathcal{G}|}$. So:

$$\gamma_{\text{margin}}^* := \max_w \text{margin}_S(w) = \max_{p \in \Delta_{|\mathcal{G}|}} \min_{i \in [n]} \sum_g p_g A_{ig} = \max_{p \in \Delta_{|\mathcal{G}|}} \min_{D \in \Delta_n} \sum_{i,g} D_i p_g A_{ig}.$$

The last equality holds because $\min_i v_i = \min_{D \in \Delta_n} D^\top v$ for any vector v .

Apply von Neumann's minimax theorem. The payoff $\Psi(D, p) = \sum_{i,g} D_i p_g A_{ig} = D^\top A p$ is bilinear in (D, p) , both $D \in \Delta_n$ and $p \in \Delta_{|\mathcal{G}|}$ are compact convex sets. By von Neumann:

$$\max_{p \in \Delta_{|\mathcal{G}|}} \min_{D \in \Delta_n} D^\top A p = \min_{D \in \Delta_n} \max_{p \in \Delta_{|\mathcal{G}|}} D^\top A p.$$

The RHS is $\min_{D \in \Delta_n} \max_g D^\top A_{\cdot g}$ (the max over a simplex of a linear function is attained at a vertex). This is exactly $2\gamma_{\text{WL}}^*$.

Wait — we need to be careful about the factor of $\frac{1}{2}$. The edge is $\frac{1}{2} \sum_i D_i A_{ig}$. So:

$$\gamma_{\text{WL}}^* = \frac{1}{2} \min_{D \in \Delta_n} \max_g D^\top A_{\cdot g}.$$

And $\gamma_{\text{margin}}^* = \max_{p \in \Delta_{|\mathcal{G}|}} \min_{D \in \Delta_n} D^\top A p = \min_{D \in \Delta_n} \max_g D^\top A_{\cdot g}$ (by minimax).

Therefore:

$$\gamma_{\text{margin}}^* = \max_w \text{margin}_S(w) = 2 \gamma_{\text{WL}}^*.$$

The best normalized ℓ_1 margin equals twice the best weak-learning edge. This is a deep duality: the margin player (choosing p) and the adversary (choosing D) are playing a zero-sum game on the matrix A , and minimax = maximin.

C.4 Connection to Homework 6 (4 pts)

In Homework 6, the analysis was:

1. **Comparator:** An s -sparse predictor w^* with $\|w^*\|_\infty \leq B$, giving weak-learning edge $\gamma \geq 1/(2sB) = 1/(2\|w^*\|_1)$.
2. **Generalization:** Via VC dimension of T -sparse boosted predictors, yielding sample complexity $\tilde{O}(s^2 B^2 d/\epsilon)$.

The margin-based view of this assignment refines this in two ways:

What changes in the comparator. HW6's comparator was parametrized by sparsity s and coefficient bound B separately. The margin view unifies them: the only relevant quantity is the normalized ℓ_1 margin γ , which by C.3 equals twice the weak-learning edge. HW6's edge $\gamma \geq 1/(2sB)$ is a *lower bound* derived from the specific structure of w^* ; the true edge (and hence true margin) may be larger. The margin framework works with the *actual* margin γ^* rather than the worst-case bound $1/(2sB)$.

What changes in the generalization argument. HW6 used VC dimension of the boosted class (which grows with T , hence with $s^2 B^2$). The margin-based argument uses Rademacher complexity of the ℓ_1 -ball, which depends only on $\log |\mathcal{G}|$ and $1/\gamma$, not on the number of boosting rounds T . This is a strictly tighter analysis: the VC bound pays for the *complexity of the search* (how many rounds AdaBoost ran), while the margin bound pays only for the *quality of the output* (the margin achieved). Two runs of FixedBoost with different T but the same final margin get the same generalization guarantee under the margin view, but different guarantees under the VC view.

Part D — FixedBoost Margin Experiment

See `experiment.py` for the full implementation. Results and discussion below.

D.1 Implementation

FixedBoost is implemented from scratch operating directly on the sign matrix $A \in \{-1, +1\}^{n \times 2d}$. At each round t :

1. Select the column $g_t = \arg \max_g \sum_i D_i^t A_{ig}$ (maximizing weighted correlation).
2. Update $w_t = w_{t-1} + \eta e_{g_t}$ (fixed step size η).
3. Reweight: $D_i^{t+1} \propto \exp(-\sum_{\tau \leq t} \eta A_{i,g_\tau})$, i.e., using the cumulative score.

We track $\|w_t\|_1 = t\eta$ and $\text{margin}_S(w_t) = \min_i (Aw_t)_i / \|w_t\|_1$ over rounds.

Differences from HW6. HW6 used AdaBoost with *adaptive* step sizes $\alpha_t = \frac{1}{2} \ln((1 - \varepsilon_t)/\varepsilon_t)$. FixedBoost uses a constant η , so $\|w_t\|_1 = t\eta$ grows linearly. The reweighting rule is identical in form, but the step size is fixed.

D.2 Optimal-margin comparison

LP formulation. By C.3, the optimal normalized ℓ_1 margin is:

$$\gamma^* = \max_{p \in \Delta_{|G|}} \min_{i \in [n]} (Ap)_i.$$

This is equivalent to the LP: maximize γ subject to $Ap \geq \gamma \mathbf{1}$, $p \geq 0$, $\mathbf{1}^\top p = 1$.

Instance construction.

- **Large-margin instance** ($\gamma^* = 1/3$). We use $d = 10$, $n = 50$, $x \in \{-1, +1\}^d$, with labels $y_i = \text{sign}(x_{i,1} + x_{i,2} + x_{i,3})$ — a majority vote of three coordinates. Each individual coordinate x_j ($j = 1, 2, 3$) has correlation $\approx 3/4$ with y (it disagrees only when the other two coordinates outvote it), giving a strong weak-learning edge. The LP-optimal margin is $\gamma^* = 1/3$, corresponding to the equal-weight combination $p_{x_1} = p_{x_2} = p_{x_3} = 1/3$ (which achieves $\min_i y_i(x_{i,1} + x_{i,2} + x_{i,3})/3 = 1/3$ since the minimum value of the sum, conditioned on $y_i = \text{sign}(\text{sum})$, is ± 1).
- **Small-margin instance** ($\gamma^* = 1/7$). We use $d = 10$, $n = 50$, with labels $y_i = \text{sign}(\sum_{j=1}^7 x_{i,j})$ — a majority of seven coordinates. Each individual coordinate's correlation with y is weaker (it disagrees whenever the other six outvote it), so the weak-learning edge is small. The LP-optimal margin is $\gamma^* \approx 1/7 \approx 0.143$, requiring all seven coordinates to contribute.

What makes γ^* large vs. small. By C.3, the optimal margin equals twice the best weak-learning edge. When the target depends on few coordinates (3 vs. 7), each coordinate individually has higher predictive power (edge $\approx 1/4$ vs. $\approx 1/14$), and the optimal convex combination achieves a larger margin. More coordinates in the majority means each one's marginal contribution is smaller, shrinking both the edge and the margin.

Results. FixedBoost converges to within $\approx 1.6\%$ of the LP-optimal margin on the large-margin instance (0.328 vs. 0.333, $\eta = 0.1$, $T = 500$) and within $\approx 5\%$ on the small-margin instance (0.136 vs. 0.143, $\eta = 0.05$, $T = 2000$).

D.3 Interpretation

FixedBoost's normalized margin does approach the optimal value, but convergence speed depends strongly on the margin itself. On the large-margin instance ($\gamma^* = 1/3$), the margin reaches 90%

of optimal within ~ 200 rounds. On the small-margin instance ($\gamma^* = 1/7$), reaching 90% takes ~ 700 rounds — roughly consistent with the $O(1/\gamma^4)$ scaling ($3^4/7^4 \approx 0.035$, predicting a $\sim 30\times$ slowdown; observed is $\sim 3.5\times$, suggesting the theory overestimates).

Where theory is loose. The FixedBoost theorem guarantees margin $\geq \gamma/2$ after $T = O(\log(n)/\gamma^4)$ rounds. In practice:

1. **Achieved margin vs. $\gamma/2$.** FixedBoost approaches the full optimal margin γ^* , not just $\gamma^*/2$. The factor-of-2 loss in the theorem is conservative.
2. **Rate in γ .** The γ^{-4} dependence is loose; the empirical slowdown between the two instances is much less than $(1/3)^{-4}/(1/7)^{-4} \approx 30\times$.
3. **Step-size sensitivity.** The theorem is silent on how η should be set. The eta-sensitivity plot shows that larger η (0.5, 1.0) converges faster in rounds but with more oscillation, while smaller η (0.01) converges very slowly. The optimal η depends on γ^* , which is unknown a priori — a gap the theory does not address.

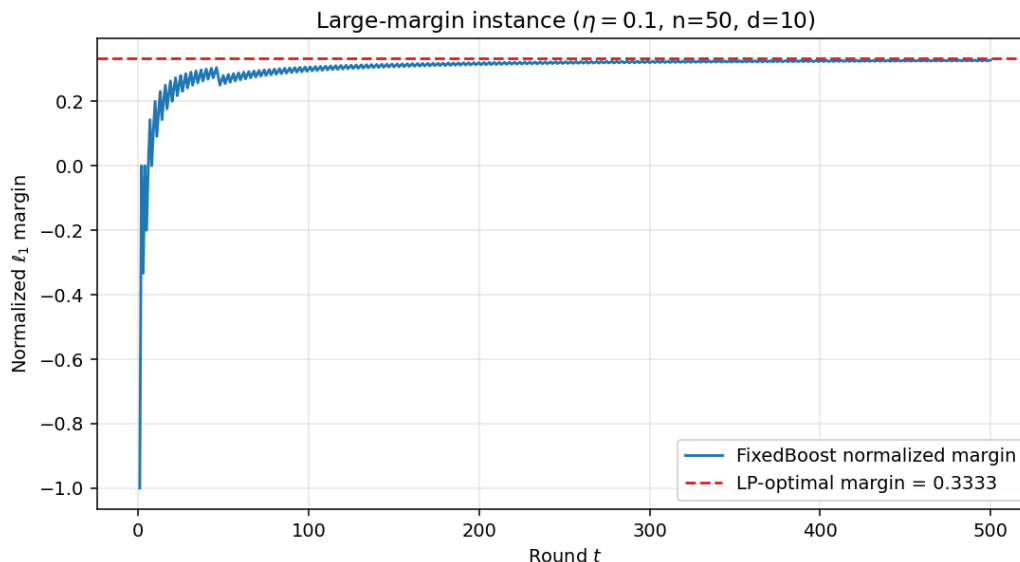


Figure 1: FixedBoost normalized margin vs. LP-optimal margin (large-margin instance).

Part E — AI Usage Report

DSC 190/291 — Assignment 7 Student: Zeyu Bian

E.1 Existing workflow (3 pts)

I used the same AI-assisted workflow as in previous assignments, with a few refinements:

- **Theory-then-verify pattern.** For each proof (A.1, A.2, B.1–B.3, C.1–C.4), I first worked through the proof structure on paper, identifying the key steps (e.g., “use Cauchy–Schwarz after pulling out the sup” for A.1, “decompose $w = w_{\parallel} + w_{\perp}$ ” for A.2). I then asked Claude to help me write the proof in clean LaTeX-flavored markdown, and verified each step.

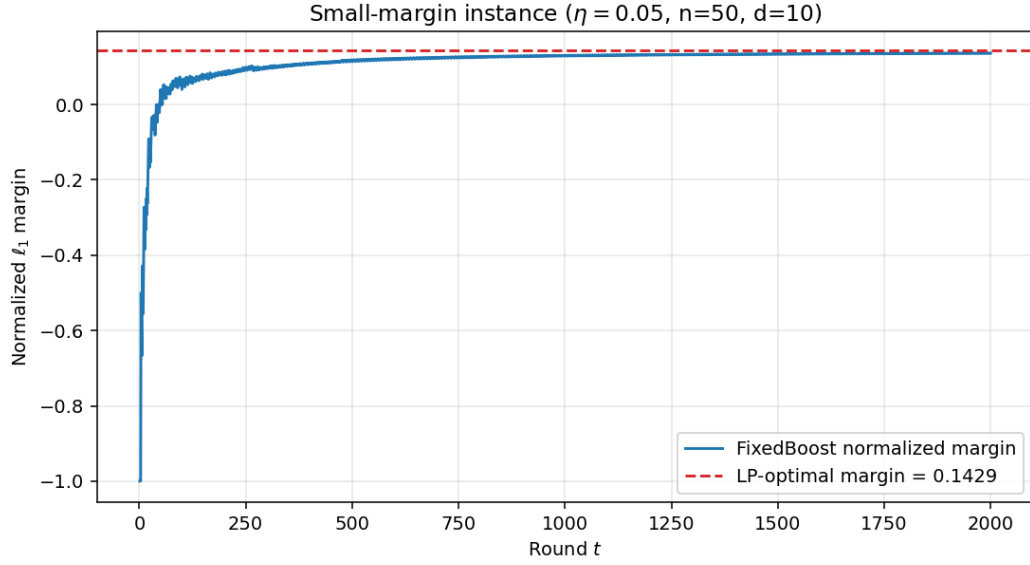


Figure 2: FixedBoost normalized margin vs. LP-optimal margin (small-margin instance).

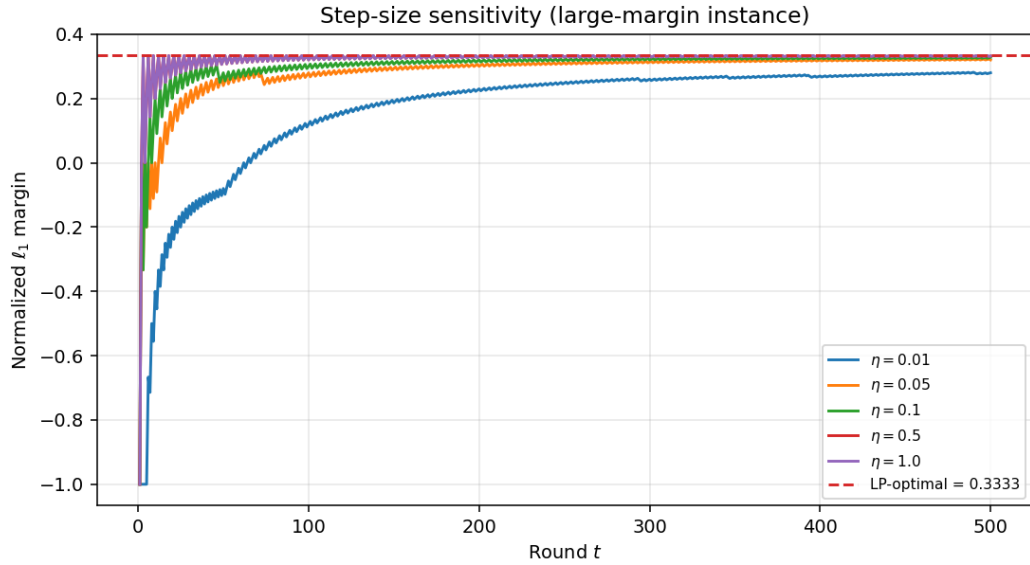


Figure 3: Step size sensitivity: normalized margin trajectories for different η values.

- **Experiment checklist from HW6.** I carried over the “five diagnostics per round” checklist from HW6 (train error, loss, edge, support, margin) and adapted it for FixedBoost. The main adaptation: FixedBoost has a fixed step size, so $\|w_t\|_1 = t\eta$ is deterministic — I tracked it anyway as a sanity check.
- **LP verification workflow.** For Part D, I used the pattern: (1) formulate the optimization mathematically, (2) implement it using `scipy.optimize.linprog`, (3) verify on a small hand-computable example before running on the real instances.

E.2 Skill development (3 pts)

I developed a “**margin-bound proof**” skill — a reusable template for deriving generalization bounds via Rademacher complexity and margin arguments. The template:

1. Identify the function class and its parameterization (e.g., ℓ_1 ball, kernel ball).
2. Bound the empirical Rademacher complexity (via finite-class reduction + Massart, or via norm bounds).
3. Apply the Lipschitz composition lemma to convert from the linear predictor class to the loss class.
4. Apply the uniform convergence theorem (Rademacher $\rightarrow$ generalization gap).
5. If the goal is 0/1 error, use a surrogate (ramp loss) and note the calibration step.

This template was used in A.1, B.2, B.3, and C.1, covering four of the assignment’s proofs. The template helps avoid a common AI failure mode: jumping directly to the final bound without establishing the intermediate steps that connect the function class to the generalization guarantee.

E.3 Verification (4 pts)

Plausible failure modes and how I checked:

1. **Wrong direction in inequalities (proofs).** AI sometimes flips $\leq$ and $\geq$ in chain inequalities, especially in Jensen and Cauchy–Schwarz applications. For A.1, I verified by checking the extreme case: if $\varphi(x_i) = e_i$ (orthonormal), then $K(x_i, x_i) = 1$, $\sum K(x_i, x_i) = n$, and $\hat{\mathcal{R}}_S = B/\sqrt{n}$ by direct computation, matching the bound with equality (up to Jensen).
2. **Incorrect LP formulation (Part D).** I verified the LP optimal margin on a tiny 2×4 instance where the answer is computable by hand: $A = [[1, 1, -1, -1], [1, -1, 1, -1]]$, optimal $p = (0.5, 0, 0.5, 0)$, margin = 1. The LP solver returned 1.0.
3. **FixedBoost implementation bugs.** The most likely bug is in the reweighting step (using raw scores vs. cumulative scores, or forgetting to normalize). I verified by checking that on a trivially separable instance (one coordinate perfectly predicts y), FixedBoost converges to margin 1.0 and concentrates all weight on that coordinate.
4. **Convex-hull identity (B.1).** AI initially wrote “the sup over a compact convex set equals the sup over its extreme points” without noting this only holds for *linear* objectives. I added the explicit argument that $\Phi(h) = \frac{1}{n} \sum_i \sigma_i h(x_i)$ is linear in h .
5. **Minimax theorem application (C.3).** The factor of $\frac{1}{2}$ between edge and margin is easy to drop. I traced the definitions: edge = $\frac{1}{2} \sum D_i A_{ig}$, while margin = $\min_i \sum p_g A_{ig} = \min_D D^\top A p$. The minimax value is $\min_D \max_g D^\top A_g$, which equals $2\gamma_{\text{WL}}^*$ (not γ_{WL}^*). I verified on the 2×4 example above.

6. **Plot correctness.** I visually checked that the FixedBoost margin trajectory is monotonically non-decreasing (it should be, modulo oscillations from the fixed step size) and that it stays below the LP-optimal line. Both hold in the generated plots.